The SUETRI-A-Team

What Should We Count? Token Accounting for the Carbon Footprint of Agentic AI

Ge Zhang, Jingru Cheng, Zitong Huang, Zhibo Dai, and Anthony R. Kovscek
Department of Energy Science & Engineering, Stanford University

Software Engineering FTE to Token Consumption

Within each stage: light · 0.30 FTE/day | medium · 0.50 FTE/day | dark · 0.70 FTE/day

Thin paths: sampled Monte Carlo days



Modeling | conditional high-load case (one engineer-day)

System boundary

Inference + Facility Electricity

1 Work to sessions

20 agent sessions/day

40 equal-effort units × q

N ~ Binomial(40, q); attempted FTE/day = N/40

2 Sessions to tokens

35.1M tokens/day

1.4M median total tokens/session

Session-token spread: lognormal σ = 0.7

3 Tokens to energy

5.94 H100 GPU-h/day

1,642 total tokens/s at 326 W board

Facility energy: 3.25 kWh/day (host × PUE)

4 Energy to emissions

U.S. HDC mix

0.325 kg CO_2e/kWh

IPCC source LCA × U.S. supply shares

Electricity LCA; EPA operational kept separate

Final outputs

1.05 kg CO_2e/day

3.25 kWh/day | 35.1M tokens/day

P10–P90: 0.72–1.46 kg/day

Attributional electricity LCA

How to read

40 candidate work units/day

- q: attempted agent share
- P10–P90: simulated day range
- LCA: electricity life cycle

Conditional high-load scenario

1 Work

40 UNITS/DAY · q = 0.50

0.50 attempted FTE/day

P10–P90 AT q = 0.50

0.40–0.60

N / 40; allocation assumption

2 Agent sessions

40 UNITS/DAY · q = 0.50

20 sessions/day

P10–P90 AT q = 0.50

16–24

N ~ Binomial(40, q)

3 Tokens

40 UNITS/DAY · q = 0.50

35.1 million tokens/day

P10–P90 AT q = 0.50

25.3–46.9

1.4M median per session

4 GPU time

40 UNITS/DAY · q = 0.50

5.94 H100 GPU-hours/day

P10–P90 AT q = 0.50

4.28–7.93

tokens / 1,642 per second

5 Electricity

40 UNITS/DAY · q = 0.50

3.25 facility kWh/day

P10–P90 AT q = 0.50

2.29–4.44

326 W board × host × PUE

6 Grid factor

40 UNITS/DAY · q = 0.50

0.325 kg CO2e/kWh (LCA)

P10–P90 AT q = 0.50

0.286–0.363

U.S. mix × IPCC factors

7 Emissions

40 UNITS/DAY · q = 0.50

1.05 kg CO2e/day (LCA)

P10–P90 AT q = 0.50

0.72–1.46

electricity × grid factor

Example: engineer-day and illustrative global scale-up

40 units/day → 20 sessions → 35.1M tokens → 5.94 H100 GPU-h → 3.25 kWh → 1.05 kg CO_2e/day

Scale-up illustration: 1.05 kg CO_2e/day × 20M professional developers × 365 days/year = 7.665 Mt CO_2e/year.

Assumes daily agent use by every developer; not a forecast or measured net FTE replacement. Day-level LCA P10–P90: 0.72–1.46 kg.

Sources

Sources: Standard Compute; local H100 tests; Guidi et al. (2026); EIA (2024); IPCC AR5; EPA eGRID; JetBrains (2025).